

AI in Healthcare Hackathon 2026

Host: Central Asian University

Start: March 26, 2026 — 08:00 (Tashkent Time, UTC+5)

Final Submission Deadline: March 27, 2026 — 20:00 (Tashkent Time)

Top 10 Teams Announcement: March 27, 2026 — 22:00

Top 10 Presentations: March 28, 2026 — 09:30 – 14:30

1. Objective

The AI in Healthcare Hackathon 2026 challenges participants to design artificial intelligence systems capable of analyzing medical biopsy images.

Participants will:

- Train AI models for **image classification** and **image segmentation**
- Work with **real medical imaging datasets**
- Apply deep learning techniques with image processing techniques for pre-processing if applicable.
- Submit predictions generated by their models
- Demonstrate their approach and solution

The competition is designed to test **technical capability, model design, and practical implementation.**

2. Dataset Access

Participants must download the dataset using the following links.

Primary dataset link:

<https://drive.google.com/drive/folders/1IBOjT4IXfNF14FtocQPGQ7D206WRKLog?usp=sharing>

QR code for the alternative link is provided below



Alternative link (if the first link does not work):

https://akfauniversity0-my.sharepoint.com/:f:/g/personal/head_cs_centralasian_uz/IgA7RA6hcu2mTIdz88oOJ5QIASk1D1vdrYEXwAYITgNKokc?e=h0o3UF

Or you can scan the QR code provided below



After downloading, **participants must verify the dataset structure and counts before starting work.**

3. Dataset Verification

After extracting the dataset, verify that the following files and folders exist.

Classification test set:

classification/test → 1276 PNG images

Classification training set:

classification/train/0 → 571 images

classification/train/1 → 974 images

classification/train/2 → 1043 images

classification/train/3 → 750 images

classification/train/4 → 814 images

classification/train/5 → 441 images

classification/train/6 → 545 images

classification/train/7 → 2136 images

classification/train/8 → 331 images

classification/train/9 → 1111 images

classification/train/10 → 899 images

classification/train/11 → 1796 images

Segmentation test images:

segmentation/testing/images → 200 images

Segmentation training data:

segmentation/training/images → 1800 images

segmentation/training/masks → 1800 masks

Segmentation validation data:

segmentation/validation/images → 400 images

segmentation/validation/masks → 400 masks

Important requirement:

Each mask in `segmentation/train/masks` must have the **same filename as the corresponding image in the images folder**.

Example:

`images/1234.png`

`masks/1234.png`

If these counts do not match, the dataset may be incomplete and should be downloaded again.

4. Dataset Description

The dataset contains **biopsy medical images**.

Classification Task

Classification Task

The classification dataset contains **twelve classes**.

Classes are labeled as numbers:

0

1

2

3

4

5

6

7

8

9

10

11

The actual disease names have been intentionally hidden.

Participants must train a model to predict the correct label.

Segmentation Task

The segmentation dataset contains images and binary masks.

Participants must train a **binary segmentation model** to detect the region of interest in the biopsy images.

The segmentation dataset is intentionally limited and requires careful model design to achieve good generalization.

5. Tasks

Participants must complete **two tasks**.

Task 1 — Image Classification

Train a model using:

```
classification/train
```

The model must predict labels for images in:

```
classification/test
```

Total test images:

1276

Allowed labels:

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11

Classification Submission File

Participants must use the provided Excel file:

```
teamName test_ground_truth.xlsx
```

Before submission:

Replace **teamName** in the filename with your actual team name.

Example:

```
Alpha test_ground_truth.xlsx
```

Excel file format:

Image_ID Label

20111	2
20112	7
...	

All 1276 images must be included

Rules:

- Image_ID must match the image filename without extension
- Label must be one of: 0, 1, 2, 3, 4, ..., 11
- All **1276 images must be included**
- No extra rows or missing rows

Task 2 — Image Segmentation

Train a segmentation model using:

```
segmentation/training/images  
segmentation/training/masks
```

You may use the following for validation

```
segmentation/validation/images  
segmentation/validation/masks
```

Generate predicted masks for images in:

```
segmentation/testing/images
```

Total test images:

200

Segmentation Mask Requirements

Predicted masks must satisfy the following:

- File format: **PNG**
- Filename must match the input image ID
- Mask size must be **exactly the same as the input image**
- Masks must be **binary**

Example:

Input image:

1476.png

Predicted mask:

1476.png

Mask Folder Structure

All masks must be stored inside a folder named exactly as your **Team Name**.

Example:

```
Alpha/  
  1476.png  
  1477.png  
  1478.png
```

6. Submission Instructions

Participants must submit the following items.

1. Classification predictions Excel file
2. Segmentation mask folder
3. Model folder containing scripts and saved models
4. Scripts that evaluate testing of segmentation and classification using the provided models

Submit the **public sharing link** through the Google Form.

Submission form:

<https://docs.google.com/forms/d/e/1FAIpQLSeGr5kulO09Rmace0yAfPs252ZapAZXMPND9t76xCUTKK2LfQ/viewform?usp=header>

Or alternatively, the following QR code may be scanned to access the Google form



7. Required Folder Structure for Submission

Participants must upload a folder named with their **team name**.

Example:

Alpha/

Inside the folder:

Alpha/

Alpha test_ground_truth.xlsx

Alpha/

1476.png

1477.png

models/

classification/

classify.py

saved classification model

requirements.txt

segmentation/

segment.py

saved segmentation model

requirements.txt

8. Script Requirements (Verification Only)

Scripts are required **only for verification purposes**.

They must perform **inference only** and must not retrain models.

Classification Script

Script name:

```
classify.py
```

Inputs:

```
directory containing test images  
submitted model trained for classification
```

Output:

```
Excel file containing Image_ID and Label
```

Segmentation Script

Script name:

```
segment.py
```

Input:

```
directory containing images  
submitted model trained for segmentation
```

Output:

```
directory containing predicted masks
```

Output masks must:

- match input filenames
- have identical image size
- be binary masks

9. Data Usage Rules

Participants may use:

- ✓ Pretrained models
- ✓ Transfer learning
- ✓ Public open-source libraries
- ✓ Public datasets for pretraining or fine-tuning

However, the following are strictly prohibited:

- ✗ Using the provided **test images for training**
- ✗ Manually labeling the test images
- ✗ Using external datasets containing the **same images as this dataset**
- ✗ Sharing predictions or models with other teams

Organizers may request training code or explanations to verify results.

Violation may lead to **disqualification**.

10. Scoring Overview (Total: 100 points)

Phase 1 — Model Performance (70 points)

- **Classification performance:** 30 points
 - Scored using **accuracy** on the test set
- **Segmentation performance:** 40 points
 - Scored using **mean IoU** on the test set

Example:

If a team achieves **83% classification accuracy** and **0.65 mean IoU** for segmentation:

$$\text{Model Score} = (0.83 \times 30) + (0.65 \times 40) = 24.9 + 26.0 = 50.9 \text{ out of } 70$$

Teams with the **top 10 model scores** will be shortlisted for the **presentation phase**.

Phase 2 — Top 10 Presentation Evaluation (30 points)

Shortlisted teams will be scored on:

- **User Interface Design:** 10 points

- **Error Handling:** 5 points
- **Innovation & Creativity:** 5 points
- **Presentation & Clarity:** 10 points

Example:

If a team scores **46.75 out of 70** in Phase 1 and **25 out of 30** in Phase 2:

Total Score = 46.75 + 25 = 71.75 out of 100

The **final ranking** is based on the **combined score**. The **top three teams** will receive prizes.

11. User Interface Requirement (Required from top 10 Teams only)

Shortlisted teams must present an interface demonstrating their AI system.

The interface should at least show:

- image upload
- classification prediction
- segmentation visualization

Allowed formats:

- web application
- desktop application
- dashboard
- prototype

The interface must clearly state:

“For research and demonstration purposes only. Not for clinical use.”

12. Presentation Guidelines (Top 10 Teams, 15 minutes maximum)

- Presentations will be conducted in **English**.
- One team member may present on behalf of the team.
- Each team must clearly explain their approach.

The presentation must include the following sections:

AI Workflow

- Overall system pipeline
- Data preprocessing steps
- Training and inference procedures

Model Design

- Models or architectures used
- Explanation of why those models were selected
- Training strategy (transfer learning, pretrained models, etc.)

Challenges and Solutions

- Key challenges encountered during development
- Methods used to address those challenges

Results

- Observations from experiments
- Performance insights
- Model behavior analysis

Interface Demonstration

Teams must demonstrate an interface showing how their AI system works. The interface should allow visualization of:

- Image input
- Model outputs
- Output visualization

Allowed formats:

- Web application
- Desktop application

- Dashboard
- Interactive prototype

Important Requirement:

The interface must clearly display the statement:

“For research and demonstration purposes only. Not for clinical use.”

13. Judging and Quality-Based Evaluation

In addition to automated performance evaluation, the judging panel will assess the overall **quality, robustness, and understanding** of the proposed solution during the presentation stage.

During the Top 10 presentations, judges may evaluate:

Technical Understanding

- The team’s understanding of the methods used
- Ability to explain the model design and workflow
- Ability to answer technical questions about implementation

Solution Quality

- Soundness of the approach
- Proper handling of data and outputs
- Reliability and stability of the system

System Demonstration

- Clarity and usability of the interface
- Correct functioning of the AI system during demonstration
- Ability to reproduce outputs using the trained model

Innovation and Engineering Quality

- Creative use of AI techniques
- Efficiency of the proposed pipeline
- Practical design considerations

Judges may ask technical questions related to the model, data processing, and implementation to verify the **authenticity and understanding** of the solution.

Teams are expected to demonstrate clear knowledge of their approach and system design.

Failure to adequately explain the system may affect the final evaluation.

14. Academic Integrity and Authenticity of Work

All submitted solutions must represent the **genuine work** and understanding of the participating team.

Teams are expected to:

- Clearly understand the methods, models, and code used in their solution
- Be able to explain their implementation and design decisions during the presentation and Q&A session
- Demonstrate how their system processes input data and generates outputs

During the evaluation stage, judges may ask technical questions about the model, data processing pipeline, implementation details, and system design to verify the authenticity of the work.

Activities that may lead to disqualification:

- Submitting work that team members cannot explain or demonstrate
- Using solutions obtained from other teams or external participants
- Copying code or models without proper understanding
- Sharing models, outputs, or solutions with other teams
- Manipulating outputs or evaluation results

If the judging panel determines that a team cannot adequately explain their solution or demonstrate understanding of their implementation, the team may be disqualified from the competition.

15. Important Notes

- Internet access is allowed
- Pretrained models are allowed
- Participants can use TensorFlow or PyTorch
- Focus on model performance, methodology, and clarity
- Judges' decisions are final

16. Deadlines & Announcements

- **Final submission deadline:** March 27, 2026 — 22:00 (Tashkent Time)
- **Top 10 teams announcement:** March 28, 2026 — 01:00
- **Top 10 presentations:** March 28, 2026 — 09:30 – 14:30

